

## **Emiscope – Smart gas emission monitoring web application for factories**

M. Niruththigan<sup>1\*</sup>, T. Tishany<sup>1</sup>, A. Abisekan<sup>1</sup>, S. Kirojan<sup>1</sup>, U. L. Kafeela<sup>1</sup>

<sup>1</sup>*Faculty of Computing and technology, University of Kelaniya, Colombo, Sri Lanka*

Air pollution caused by harmful gas emissions from factories is a serious problem. When gases like carbon monoxide (CO) are released into the air, it affects both the environment and people's health. Many factories struggle to keep track of their gas emissions properly. Our group project, EMISCOPE, is a smart web-based application that helps monitor CO gas emissions in real time and notifies the responsible people if levels go beyond safe limits. The system mainly uses the MQ-7 gas sensor and an ESP32 microcontroller to detect CO levels. When the gas level increases, the data is sent to the Firebase Realtime Database. Then it is displayed live on a dashboard built with ReactJS and styled using Tailwind CSS. We also used Chart.js to show gas level changes using graphs, and Email.js to send alert messages to factory owners and government admins automatically. Firebase Cloud Functions are used to check the gas levels and trigger these alerts. Firebase Authentication is also used to manage login access for users. The system is easy to use and supports two types of users: factory owners and government admins. It is low-cost, accurate, and helps factories stay within legal emission limits. In future, we plan to add support for other gas sensors such as CO<sub>2</sub> and NO<sub>2</sub>, add a mobile app version, and expand the system to be used by more industries. This project is a good example of using technology to support environmental safety. It makes gas monitoring simple and automatic.

**Keywords:** Real time monitoring, Carbon Monoxide, ESP32, web application, gas emissions

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**E-mail:** nirutht-ct21081@stu.kln.ac.lk